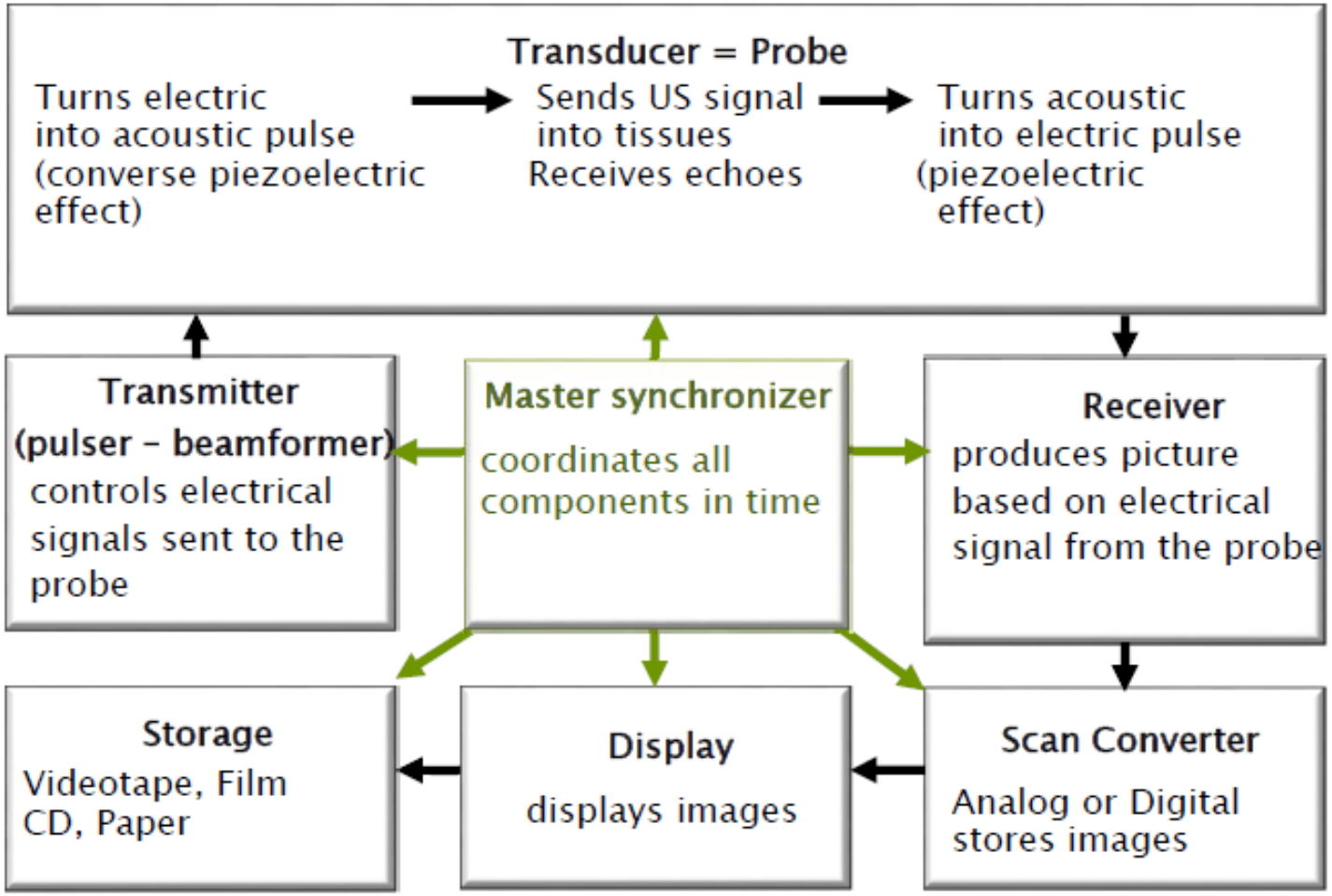
MIT 232 Lecture 1

2D Instrumentation

System Components

- Transducer – transforms electrical energy into acoustic energy and vice versa
- Pulser and Beam Former – controls electric signals sent to the transducers
- Receiver – Transforms the electrical signals from the transducer into a form
- that can be displayed
- Display – Presents the processed data as an image or plot
- Storage- archives the data from an ultrasound study
- Master Synchronizer – maintains the proper timing of the system's components

Interaction of System Components



Pulser/Beamformer

The Pulser and Beamformer refer to electronic elements that determine the pulse duration, pulse frequency and shape the beam using phased time delays etc. The process of using multiple sound waves to steer and focus a beam of sound is commonly referred to as beamforming. The functions of a beamformer include generating transmit timing and possible apodization during transmit, supplying the time delays and signal processing during receive, and supplying apodization and summing of delayed echoes.

- For transmit and receive beams, beamformer creates
  - the appropriate phase delay
  - pulse sequencing
- Apodization technique is controls
  - which elements in the array are active and which ones are not
  - how much stimulation (power) each element gets

Pulser Modes and Controls

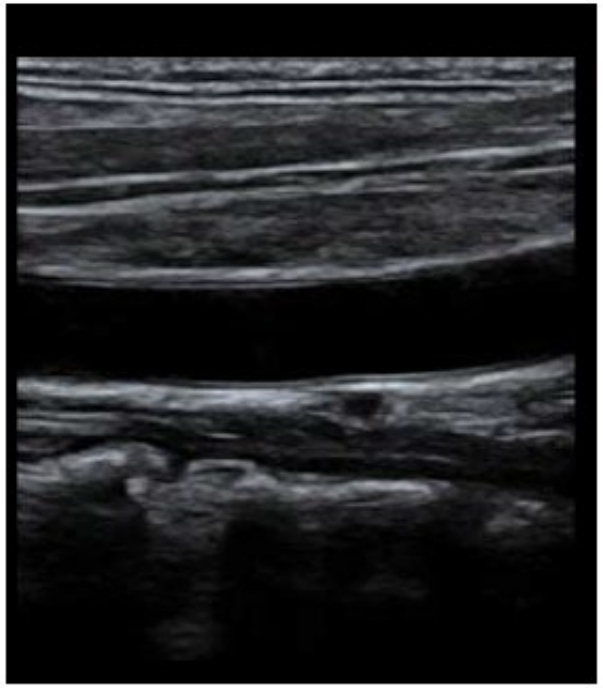
- CW
  pulser produces continuous electrical signal that is converted into continuous US wave
- PW
  pulser creates an electrical "spike" that the transducer crystal later transfers into a single US pulse.

If the transducer has many crystals (array), each crystal will receive its own "spike". Therefore, several "spikes" are needed from the pulser to generate one acoustic pulse. Timing between the "spikes" determines focusing and steering of the acoustic pulse.

Controls:

- *Depth*
- *Steer*
- *Focus*
- *Number of focal points (Focal Number)*
- *Line density*
- *Image shape and size (FOV, ExFOV)*
- *Acoustic power (output gain)*

Linear Steer



Steer off

Steer on

Focus

Focus shallower

Focus deeper

Focal Number

One Focal Point

Four Focal Points

Line Density

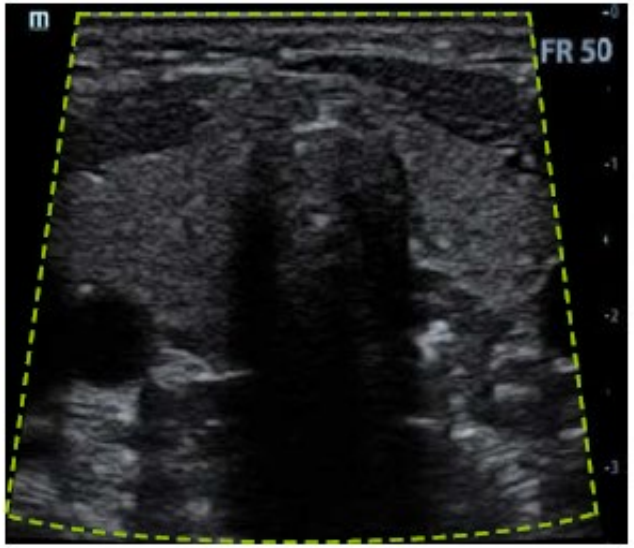
low line density

high line density

Image Shape

- Linear Transducer

Off

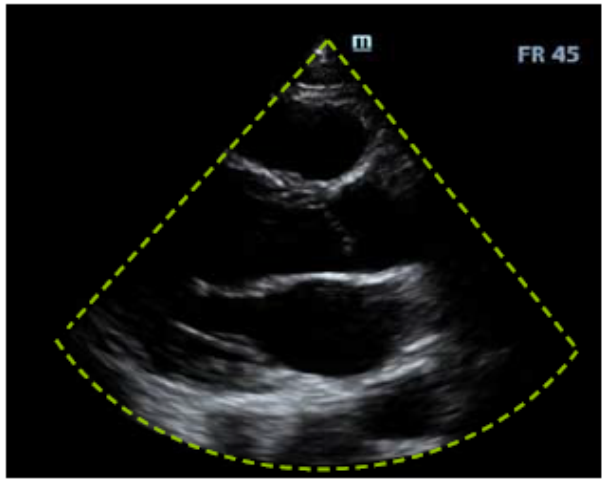


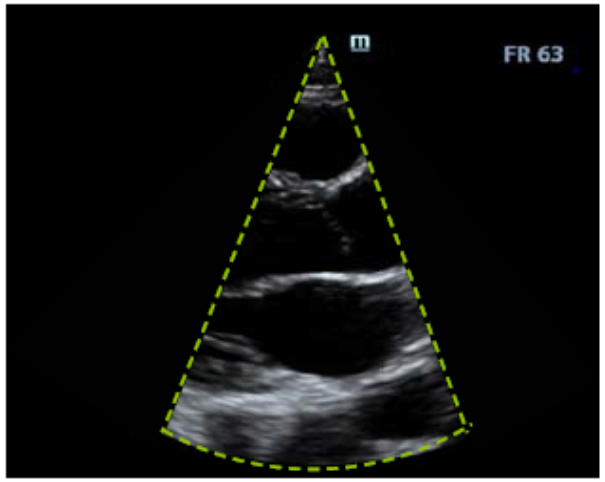
On

ExFOV (Extended Field of View)

Image Shape

- Sector Transducer





FOV (Field of View)

Acoustic Power

- Pulser receives timed signal from master synchronizer and produces voltage that is transmitted to the probe.

$$Voltage \sim Power \sim Intensity\ of\ US\ wave$$

- Therefore, transducer output is regulated by excitation voltage from the pulser and can be controlled by sonographer.
- **Controls:** *acoustic power (transducer output, output gain, pulser power, energy output, transmitter power)*

Acoustic Power

Acoustic Power low

Acoustic Power high

Mechanical and Thermal Index

These are indicators of possible bieffects (cavitation and heat). Values are typically indicated on The display

- Output power may be expressed in
  - % of maximum power or dB, where 0dB is maximum
  - Mechanical Index, MI

$$MI = \frac{peak\ rarefactional\ Pressure}{\sqrt{f}}$$

  - Thermal Index, TI for different tissues

$$TI = \frac{Output\ Power}{P\ to\ cause\ heating\ by\ 1^{o}C}$$

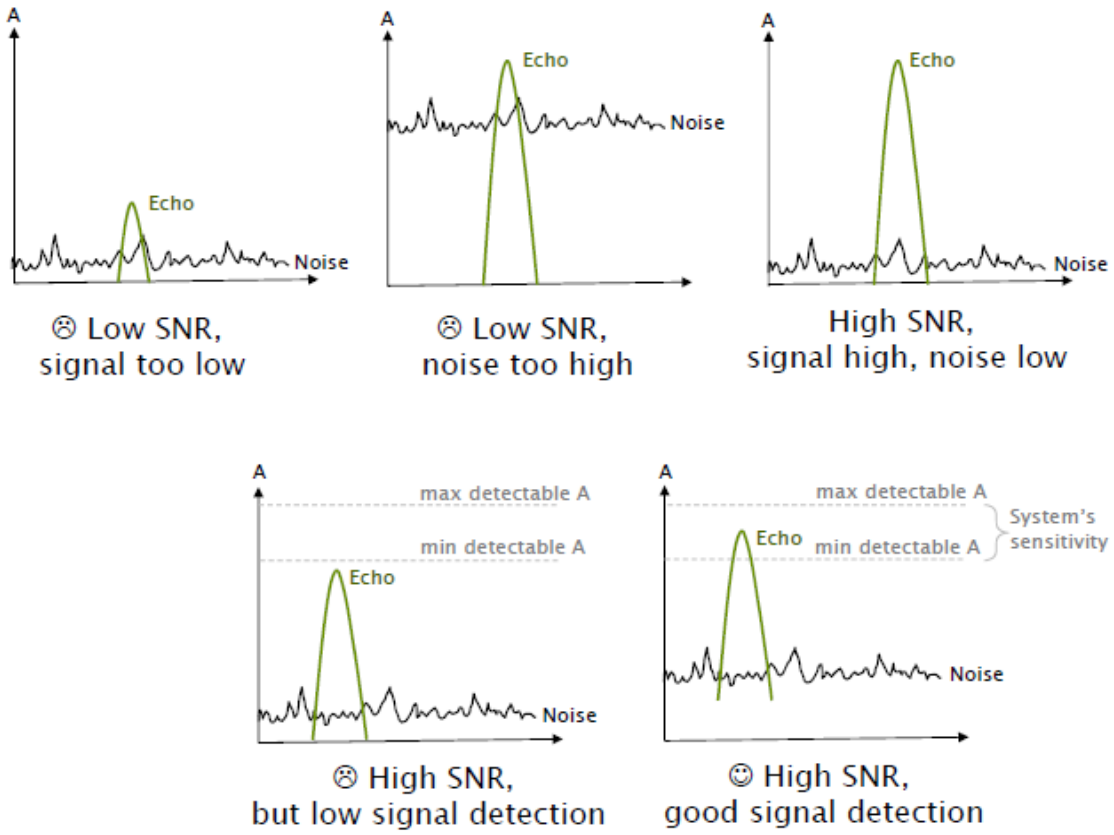
Noise

- Waves generated by different structures interact constructively and destructively, producing low amplitude echoes (noise).
- Signal-to-Noise Ratio

$$SNR = \frac{A_{echoes}}{A_{noise}}$$

- With higher amplitude signal, SNR is high.

Signal and Noise



SNR dilemmas

- Dilemma:
  - High output power is beneficial to increase SNR ☺
  - High output power increases patient's exposure ☹
- Dilemma
  - Long pulse with low amplitude will also deliver high energy (like high power, only over longer time) to improve SNR ☺
  - Long pulse decreases LARD and creates narrow bandwidth ☹